Morse Lemma. (Normal forms) differential topology enables one to analyze the topology of a manifold by studying differentiate functions on that manifold.

Diffeomorphisms 微分同胚　　光滑：无限可微.

bijective 双射 mapping : $F: U \to V$, F^{-1} denoting the inverse mapping.

定义：

F is C^k-diffeomorphism $(k \geq 1)$ if $F \in C^k(U, V)$ and $F^{-1} \in C^k(V, U)$

↓ Continuous differentiable.　C^1：一阶连续可微.　C^0：连续.

导数存在且连续

-morphism：形式、结构、

isomorphism 同构

homeomorphism 同胚

diffeomorphism 微分同胚

Optimality Conditions 最优性条件

优化问题中：如果 x^* 是局部最优解

它必须满足什么条件？

本质："一个解是最优的，需要满足某些方程/条件"

{ 极值问题的必要条件
拉格朗日乘子法.

思想：calculus of variations 变分法

the method of Lagrange multipliers：寻找 local maxima and minima.

(Multivariable Calculus)

例子：maximize $f(x, y) = xy$

subject to $3x + y = 720$　　$g(x, y) = 3x + y - 720 = 0$

$$F(x, y, \lambda) = f(x, y) + \lambda g(x, y) = xy + \lambda(3x + y - 720)$$

$$F_x(x, y, \lambda) = 0 \Rightarrow y + 3\lambda = 0$$

$$F_y(x, y, \lambda) = 0 \Rightarrow x + \lambda = 0 \qquad \Rightarrow \lambda = -120 \quad x = 120, \; y = 360.$$

$$F_\lambda(x, y, \lambda) = 0 \Rightarrow 3x + y - 720 = 0$$

The solutions are critical points of F.

PCA 优化最大方差：$\max u^T S u$

subject to $\|u\| = 1$

$$F(u, \lambda) = u^T S u + \lambda(u^T u - 1)$$

$$F_u = u^T S + \lambda u^T = 0 \qquad u^T S = -\lambda u^T \Rightarrow S^T u = -\lambda u$$

对称 $\Rightarrow S u = -\lambda u$

$$F_\lambda = u^T u - 1 = 0$$

eigenvalue.

Lagrange Multipliers：把"有约束问题"变成：

$$L(x, \lambda) \equiv f(x) + \lambda g(x)$$

Known as the Lagrangian function. or just Lagrangian.

把带约束的优化问题，重新表示成一个无约束的函数（但引入了额外变量——乘子）

然后要求：

$$\nabla_x L(x, \lambda) = 0, \quad g(x) = 0$$

KKT 本质上是 Lagrange Multipliers 对不等式约束的扩展

问题：为什么 Lagrange Multipliers 可以"自动处理约束"？

切空间　tangent space

对偶性　duality.

梯度 = 法向量

The basic idea is to convert a constrained problem into a form such that the derivative test of an unconstrained problem can still be applied.

Lagrangian 作为构造对偶性的核心工具 (Duality)

Duality：通过另一个视角（对偶问题）来估计原问题的最优值，并找到这个估计的上限或下限.

Lagrangian是工具，用于构造对偶问题.

Second normal form: quadratic function

~~SNF~~ 和数据库的概念不是同一回事.

level set 定义：$L_c(f) = \{(x_1, \dots, x_n) \mid f(x_1, \dots, x_n) = c\}$.

Differential topology / geometry

foundation: smooth manifolds

Morse Lemma: 微积分 → 变分法 → 拓扑.

起点：极值与二阶近似：为什么 critical point 附近函数"像二次型"？ Quadratic form

Quadratic approximations: The Taylor polynomial of order 2 generated by a twice-differentiable function $f(x)$ at $x=a$ is called the quadratic approximation of f at $x=a$.

linearization: Taylor polynomial of order 1.

题目：$f(x) = \ln(\cos x)$ $\quad f(x) = e^{\sin x}$ $\quad f(x) = 1/\sqrt{1-x^2}$ $\quad f(x) = \cosh x$ $\quad f(x) = \sin x$ $\quad f(x) = \tan x$

二阶近似本质上是用局部的二次函数 (quadratic form) 去替代复杂函数.

用途：判断极值类型（极小/极大/鞍点）.

$$\left.\begin{aligned} & f(x) = \ln(\cos x) \\ & f'(x) = \frac{(\cos x)'}{\cos x} = \frac{-\sin x}{\cos x} = -\tan x \qquad f'(0) = 0 \\ & f''(x) = -(\sec^2 x) \qquad f''(0) = -1 \end{aligned}\right\} \text{二阶近似}$$

$$f(x) \approx -\frac{1}{2}x^2$$

The careful analysis of non-degenerate (Hessian is non-degenerate) functions, so called Morse functions, is known as Morse Theory.

Morse Theory 最深的价值：用局部二阶信息，推断全局结构

（最神奇的地方，只研究少数几个临界点，就能知道整个空间长什么样）

Critical points：临界点，满足 $f'(x) = 0$ 或 $\nabla f(x) = 0$

题目：Does $f(x) = 2e^x/(1+x^2)$ have an absolute minimum value? An absolute maximum? If so, find them or give reasons why they fail to exist. List all critical points of f.

高法见刚求导

Quotient rule: $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$

$$f'(x) = \frac{2e^x(1+x^2) - 2e^x \cdot 2x}{(1+x^2)^2} = \frac{2e^x(x-1)^2}{(1+x^2)^2}$$

$f'(x) = 0 \Rightarrow x = 1$ $\qquad f'(x) \geq 0$ 单调增

$x = 1$ 是 critical point，但它并非极值（非极大/非极小），而是 flat stationary point.